USN: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

[23CSE306]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

ARTIFICIAL INTELLIGENCE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Define Rational Agent.	L1	CO1	**02**
	b.	Draw the figure to illustrate the agent and environment.	L2	CO2	**02**
	c.	Define a problem using the four basic components.	L1	CO3	**02**
	d.	What is Declarative Knowledge?	L2	CO4	**02**
	e.	Define Classical Planning.	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Define Artificial Intelligence and describe its significance in modern technology.	L2	CO1	**05**
		OR			
	b.	Explain the concept of "acting rationally" and its importance in AI	L2	CO1	**05**
3.	**a.**	Describe how an agent's environment affects its behavior.	L2	CO2	**05**
		OR			
	b.	Discuss the structure of a utility-based agent with an example.	L2	CO2	**05**
4.	**a.**	What are problem-solving agents? illustrate the term with an example.	L3	CO3	**05**
		OR			
	b.	Apply constraint satisfaction to a sample problem and explain the solution structure.	L3	CO3	**05**
5.	**a.**	Demonstrate the syntax and semantics of first-order logic.	L3	CO4	**05**
		OR			
	b.	Show how backward chaining can be used to make inferences in a logical agent.	L3	CO4	**05**
6.	**a.**	Explain the purpose of heuristics in planning algorithms.	L2	CO5	**05**
		OR			
	b.	Describe how planning can be adapted in non-deterministic domains with an example.	L2	CO5	**05**
7.	**a.**	Illustrate the concept of heuristic functions in planning search algorithms.	L3	CO6	**05**
		OR			
	b.	Define the WUMPUS world and plan the solution using any planning technique.	L3	CO6	**05**

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

		PART-C			6 X 10=60
8.	**a.**	Describe the foundations of AI and the contributions of early pioneers.	L2	CO1	**10**
		OR			
	b.	Explain the AI history how neural networks have evolved and returned in AI applications.	L2	CO1	**10**
9.	**a.**	Compare the differences between goal-based and utility-based agents.	L3	CO2	**10**
		OR			
	b.	Give a scenario to determine which agent structure would best fit and explain why.	L3	CO2	**10**
10.	**a.**	Explain the informed search strategies with suitable examples.	L2	CO3	**10**
		OR			
	b.	Describe uninformed search strategies to solve a specific search problem.	L2	CO3	**10**
11.	**a.**	Illustrate the differences between propositional logic and first-order logic in AI.	L3	CO4	**10**
		OR			
	b.	Employ forward chaining to solve a given inference problem (Assume the problem).	L3	CO4	**10**
12.	**a.**	Demonstrate the significance of time, schedules, and resources in planning systems.	L3	CO6	**10**
		OR			
	b.	Use a classical planning algorithm to any scenario and evaluate the steps involved.	L3	CO6	**10**
13.	**a.**	What are knowledge-based agents? Describe their characteristics.	L2	CO5	**10**
		OR			
	b.	Explain the statement "Inference system is complete" with suitable example.	L2	CO5	**10**